

VLSI VL82C420

SCAMP IV Multiplex System Controller

Reconstructed Technical Reference

No datasheet for the VL82C420 was ever released to the public, and no databook covering the SCAMP IV chipset (VL82C420 system controller + VL82C144 peripheral chip + optional VL82C146 ExCA controller) was ever archived. SCAMP IV was announced by VLSI Technology's Portable Systems Division (Tempe, Arizona) in June 1993 and shipped in volume that October, just as VLSI was leaving the PC-chipset business.

This document reconstructs the chip's architecture and bus interface from the primary surviving engineering source: **US Patent 5,793,990**, "Multiplex address/data bus with multiplex system controller and method therefor," filed **11 June 1993** by the VL82C420 designers **James J. Jirgal, David R. Evoy and Walter H. Potts** and assigned to **VLSI Technology, Inc.** The patent's "multiplex system controller" is the VL82C420; its "Multiplexed Local (ML) Bus" is the proprietary interconnect VLSI advertised for SCAMP IV. The block diagrams on the following pages are faithfully redrawn from the patent figures; the signal, format and timing tables are transcribed from its text.

Family	SCAMP IV (VL82C420 / VL82C144 / VL82C146)
Role	System controller (CPU/memory/ML-bus + DMA)
Part / package	VL82C420FC4 (quad flat pack)
Process	0.8 µm CMOS, mixed 3.3 V / 5 V operation
CPU support	Power-managed Intel 486-class up to 33 MHz, incl. clock-doublers
Memory	Up to 32 MB
Interconnect	VLSI proprietary Multiplexed Local (ML) Bus
Announced / volume	June 1993 / October 1993
Used in	IBM ThinkPad 230Cs (Type 2432, Japan); AST Ascentia 486 notebooks

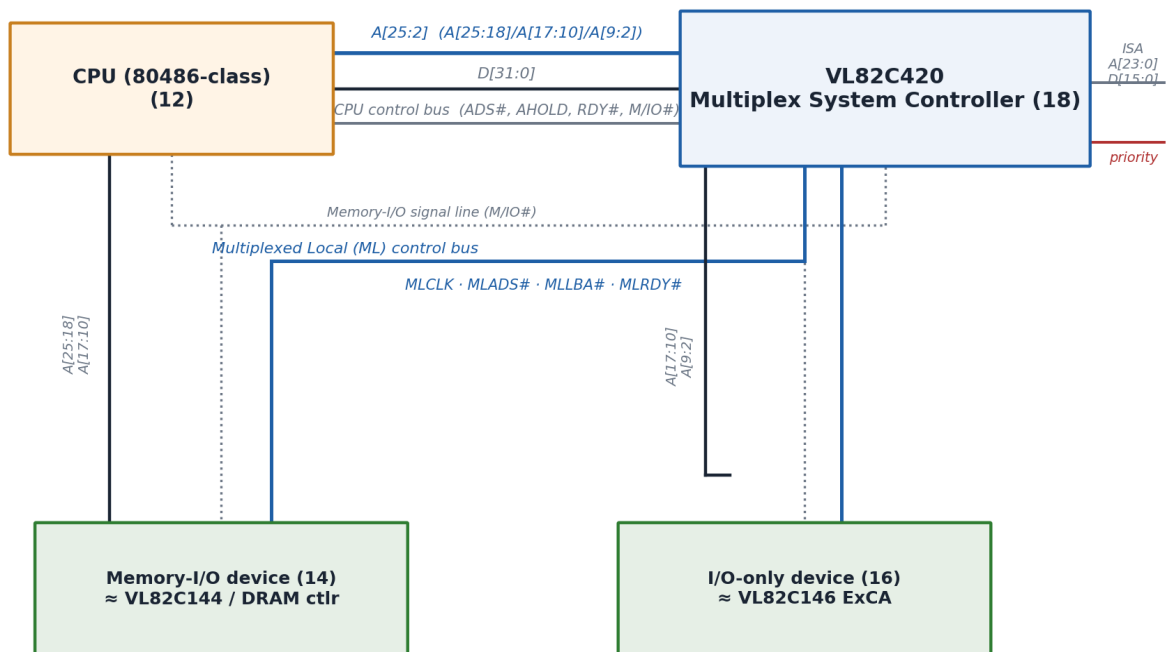
Source: US 5,793,990 A / WO 1994/029797 A1 (VLSI Technology, Inc.). Trade specs from Computergram/Tech Monitor, 17 June 1993. Use notes confirmed via The Retro Web and MacDat.

System architecture (FIG. 1)

The VL82C420 sits between the CPU and the companion devices. In a local-bus arrangement the CPU's 32-bit address bus A[25:2] is split into three groups — A[25:18], A[17:10], A[9:2] — and shared with the controller and the ML-bus devices. The controller couples to the CPU over A[25:2], D[31:0] and the CPU control bus, and drives the ML control bus (MLCLK, MLADS#, MLLBA#, MLRDY#) to both a memory-I/O device (the VL82C144 class) and an I/O-only device (the VL82C146 ExCA class). It also presents an ISA bus (A[23:0], D[15:0]) and a priority line.

VL82C420 — SCAMP IV System (local-bus architecture)

redrawn from US 5,793,990 FIG. 1



The controller tri-states the CPU address bus (AHOLD) and time-shares A[25:2] to send two 16-bit address groups then one 16-bit data group over the same wires — cutting pin count on the companion chips.

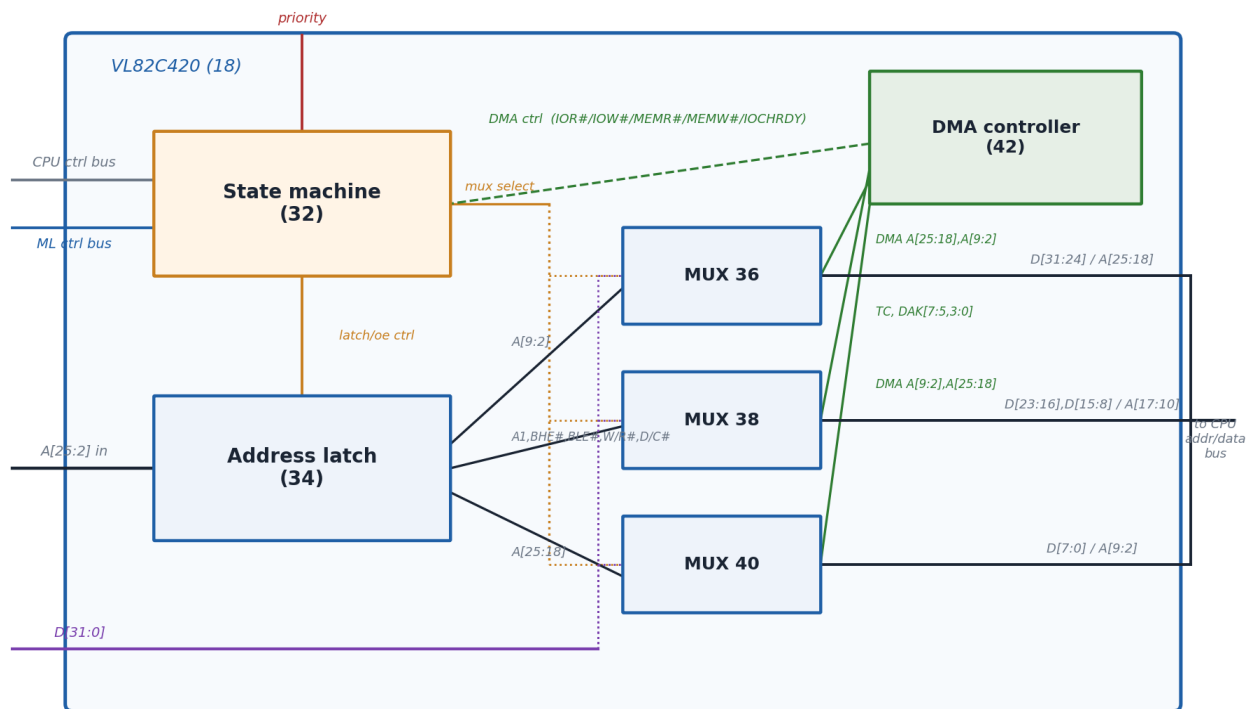
FIG. 1 — SCAMP IV local-bus system, redrawn from US 5,793,990.

Internal block diagram (FIG. 14)

Inside the VL82C420, a **state machine (32)** controls the chip from the ML control bus, the CPU control bus and a priority input. It drives an **address latch (34)** that temporarily stores the incoming address, and steers three **multiplexers (36, 38, 40)**. The latch feeds A[9:2] to MUX 36; A1, BHE#, BLE#, W/R#, D/C# to MUX 38; and A[25:18] to MUX 40. The data bus D[31:0] reaches all three. A **DMA controller (42)** feeds DMA addresses and TC/DAK acknowledges into the muxes and exchanges control with the state machine. The mux outputs drive the shared CPU address/data lines: MUX 36 → D[31:24]/A[25:18]; MUX 38 → D[23:16],D[15:8]/A[17:10]; MUX 40 → D[7:0]/A[9:2].

VL82C420 — Multiplex System Controller internal block diagram

redrawn from US 5,793,990 FIG. 14



State machine 32 drives the address latch 34 and steers multiplexers 36/38/40 to place two address groups then one data group onto the shared CPU address/data lines; DMA controller 42 feeds DMA addresses/acknowledges into the muxes.

FIG. 14 — VL82C420 multiplex system controller internals, redrawn from US 5,793,990.

Multiplexed Local (ML) Bus signals

Signal	Description (transcribed from US 5,793,990)
MLCLK	Multiplex bus clock. A 1X clock driven synchronously to the CPU clock, but on a separate line so the CPU clock and MLCLK can be gated independently for power conservation.
MLADS#	Multiplex address strobe (low-true). Asserted by the controller when it is driving valid address or data information on the CPU's address lines.
MLLBA#	Multiplex device local-bus-access (low-true). A target device asserts it when it positively decodes a valid address. During reads it also signals which CPU address lines carry the data: device on A[17:2] if MLLBA# asserted during MLRDY#, on A[25:10] if negated.
MLRDY#	Multiplex ready. A device asserts it to signal write-data accepted or read-data valid; the controller may assert it to terminate a cycle when no device acknowledges the presented address.

CPU-side and DMA signals

Signal	Description (transcribed from US 5,793,990)
ADS#	CPU address strobe; asserted marks the start of a CPU or DMA memory cycle.
M/IO#	CPU memory/I/O status; high with ADS#/MLADS# low = memory space, low = I/O space.
AHOLD	Address hold. The controller asserts it to tri-state the CPU's address bus; the CPU releases the bus on the next rising clock edge.
RDY#	CPU ready. The controller asserts it to tell the CPU that read data is valid or write data was accepted (also for DMA).
LBA#	Local-bus access from a VL-bus compatible device, asserted the clock after ADS# when it decodes the current address.
IOR# / IOW#	DMA-asserted I/O read / write strobes.
MEMR# / MEMW#	DMA-asserted memory read / write strobes.
IOCHRDY	I/O channel ready; negated by the state machine to make the DMA controller delay the current cycle.
TC, DAK[7:5,3:0]	DMA terminal count and channel acknowledges, multiplexed onto the bus via MUX 38.

16-bit address/data formats (FIG. 3A / 3B)

Each multiplex cycle transfers three 16-bit groups over the shared lines: two address groups then one data group (D[15:0]).

Memory-I/O devices (Table 1, FIG. 3A): attached to A[25:10], so the first 16-bit group (A25–A10) selects the device with 1 KB granularity — memory must therefore not be split between two devices in blocks smaller than 1 KB. The second group carries A9–A2 plus A1, BHE#, BLE# (byte selects), W/R# (write/read) and D/C# (data/code); two 'R' bits and the LSB are reserved. These devices accept both 16-bit memory and I/O cycles and respond early because they sit at higher addresses.

I/O-only devices (Table 2, FIG. 3B): attached to A[9:2] and A[17:10], decoding the x86 I/O space to 4-byte granularity on the first group. If a device owns the full 4-byte space it may return an early acknowledge; if it owns only part, it waits for the second group before asserting MLLBA#. These devices accept 16-bit I/O cycles only and reside in the low 64 KB of I/O space.

Operation & bus cycles

The controller translates ordinary x86 memory/I/O cycles into 16-bit ML cycles in which address and data share A[25:2]. On a 486-class bus it asserts AHOLD to tri-state the CPU address bus, drives the first address group with MLADS#, then (after the device asserts MLLBA#) the second address group, then the 16-bit data group; MLRDY# completes the transfer and AHOLD is released. A priority input lets a higher-priority device pre-empt and terminate an ML cycle. The patent details cycles for: I/O read from a high device (FIG. 5) and low device (FIG. 6), I/O write (FIG. 7), terminated I/O (FIG. 8), memory read (FIG. 9), memory write (FIG. 10), terminated memory (FIG. 11), and multiplex DMA read/write to multiplex memory (FIG. 12/13).

Provenance & how to obtain the originals

Primary source (this reconstruction):

US 5,793,990 A — patents.google.com/patent/US5793990A/en | PDF: patentimages.storage.googleapis.com/89/6b/d8/84459ba2e43015/US5793990.pdf | PCT: WO 1994/029797 A1

Companion VLSI / SCAMP IV patents (rest of the chipset):

- US 5,715,467 A — Event-driven power management control circuit (SCAMP IV power management).
- US 5,958,055 A — Power management system for a computer.
- US 5,561,772 A — Expansion bus replicating an internal bus as an external bus with logical interrupts.
- US 5,805,901 A — Mapping interrupt requests in a high-speed CPU interconnect bus.
- US 5,655,142 A — High-performance derived local bus.
- US 5,652,847 A — Multiplexing data and a portion of an address on a bus.

Closest VLSI-published datasheets that do exist (predecessor architecture, on bitsavers):

- SCAMP II — VL82C316 / VL82C323 data manual (Oct 1992).
- VL82C114 Combination I/O (Mar 1993) — closest sibling to the VL82C144.
- SCAMP — VL82C310 / 311 / 311L data manual (Jan 1992).

Board-level schematics: none are public. The two machines known to use the VL82C420 — the IBM ThinkPad 230Cs (Japan-only; only a Japanese brochure survives) and AST Ascentia 486 notebooks — have no digitized service manual or schematic. Recovering one would require scanning a paper IBM/AST maintenance manual.

Reconstructed reference compiled from public patent records. Diagrams are clean redraws of the patent figures, not the original VLSI artwork. For verbatim figures and claims, consult US 5,793,990 directly.